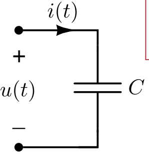
$$u(t) = \sqrt{2}U\cos(\omega t + \varphi_u) = \Re e[\sqrt{2}e^{j\omega t}\mathcal{U}]$$

$$i(t) = \sqrt{2}I\cos(\omega t + \varphi_i) = \Re e[\sqrt{2}e^{j\omega t}\mathcal{I}]$$



$$\boxed{i(t) = C \cdot \frac{du(t)}{dt}}$$

$$\Downarrow$$

$$\Re e[\sqrt{2}e^{j\omega t}\mathcal{I}] = \Re e[\sqrt{2} \cdot j\omega e^{j\omega t}C\mathcal{U}]$$

$$\Downarrow$$

$$\Re e[\sqrt{2}e^{j\omega t}\left\{\mathcal{I} - j\omega C \cdot \mathcal{U}\right\}] = 0 \quad \Rightarrow \quad \boxed{\mathcal{U} = \frac{-j}{\omega C} \cdot \mathcal{I}}$$